

DO-IT-YOURSELF: PROJECT

charge a 3.3V lithium-ion supercapacitor and operate effectively for prototype applications.

Second design - LTC3225 supercapacitor charger.

The second design is the recommended implementation, as it uses the LTC3225, a dedicated supercapacitor charging IC.

The LTC3225 enables efficient charging, improved control, and higher performance compared to battery charger ICs. It incorporates internal charge-pump circuitry and provides automatic balancing for series-connected capacitors.

In this configuration, the LTC3225 is connected to the power input connector (U4), from which it receives power and charges the supercapacitor bank. Once charged, the stored energy is supplied to a boost converter stage using the MT3608, which generates an adjustable output voltage between 5V and 25V DC.

To enable automatic backup operation, the system integrates the MAX40200 ideal diode IC. This component ensures low-loss power-path control, allowing current to flow from the supercapacitor to the load only when required.

During normal operation, the IoT device is powered by the primary source. If the main supply fails, the MAX40200 automatically switches to the supercapacitor, ensuring uninterrupted operation for a short duration. This allows the device to complete essential final tasks such as saving data or transmitting status information before shutdown.

This circuit has been partially tested. Complete validation and performance data will be presented in a subsequent revision once all components are available.

Testing

For testing the first batteryless power bank, the supercapacitors are charged using the USB port on the TP4056 module. Once fully charged, the load is connected to the MT3608, and the output voltage is set between 5V and 25V DC using the potentiometer.

For the second circuit, the supercapacitor is charged using an external DC power source, and the IoT device is connected to the output pin connector (U8). During normal operation, the device is powered by the main supply while the supercapacitor charges.

When the primary power source fails, the system automatically switches to the supercapacitor. This enables the IoT device to perform critical last-minute operations, such as sending alerts, saving configuration settings, and logging essential data before shutdown. **EFY**

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